

Data Structures



Lecture 2 Array Functions

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Refresh your memory

1. `for i in range (stop)`
2. `for i in range (start, stop)`
3. `for i in range (start, stop, step)`
4. Negative step?
5. Negative stop?

Array - Accessing/Changing an Element

Suppose you have a 1D array named *student_names*. If you want get the 5th student from the array then you write:

```
fifth_student = student_names[5]
```

To change the value at 5th index and set it to 'John Doe', you write:

```
student_names[5] = 'John Doe'
```

Array - Initializing

```
array = [None] * length
```

Array - Functions (Iteration)

```
1 def iteration(source):
2     for i in range(len(source)):
3         print(source[i])
4
5 def reverseIteration(source):
6     for i in range(len(source) - 1, -1, -1):
7         print(source[i])
```

Array - Functions (Copy Array)

```
1 def copyArray(source):  
2     newArray = [None] * len(source)  
3     for i in range(len(source)):  
4         newArray[i] = source[i]  
5     return newArray
```

Array - Functions (Resize Array)

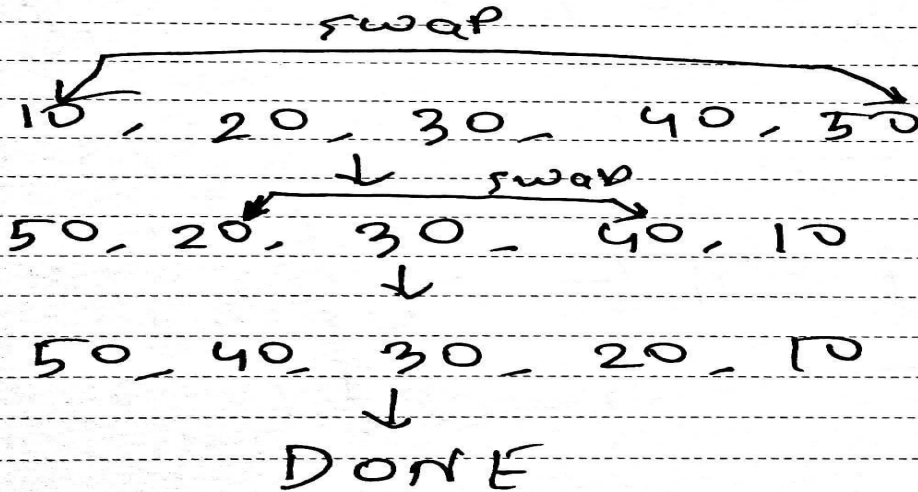
```
1 def resizeArray(oldArray, newCapacity):  
2     newArray = [None] * newCapacity  
3     for i in range(len(oldArray)):  
4         newArray[i] = oldArray[i]  
5     return newArray
```

Array - Functions (Reverse Array)

```
1 def reverseArrayOutOfPlace(arr):  
2     revArr = [None] * len(arr)  
3     i = 0  
4     j = len(arr) - 1  
5     while i < len(arr):  
6         revArr[i] = arr[j]  
7         i += 1  
8         j -= 1  
9     return revArr
```


Array - Functions (Reverse Array) (Efficient)

In place Reverse:



Array - Functions (Reverse Array) (Efficient)

```
1 def revArrInPlace(arr):  
2     i = 0  
3     j = len(arr) - 1  
4     while i < j:  
5         temp = arr[i]  
6         arr[i] = arr[j]  
7         arr[j] = temp  
8         i += 1  
9         j -= 1
```

Array - Functions (Shift Left)

```
1 def shiftLeft(arr):  
2     for i in range(1, len(arr)):  
3         arr[i-1] = arr[i]  
4     arr[len(arr) - 1] = None  
5     return arr
```

Array - Functions (Shift Right)

```
1 def shiftRight(arr):  
2     for i in range(len(arr) - 1, 0, -1):  
3         arr[i] = arr[i - 1]  
4     arr[0] = None  
5     return arr
```

Array - Functions (Rotate Left)

```
1 def rotateLeft(arr):  
2     temp = arr[0]  
3     for i in range(1, len(arr)):  
4         arr[i-1] = arr[i]  
5     arr[len(arr) - 1] = temp  
6     return arr
```

Array - Functions (Rotate Right)

```
1 def rotateRight(arr):  
2     temp = arr[len(arr) - 1]  
3     for i in range(len(arr) - 1, 0, -1):  
4         arr[i] = arr[i - 1]  
5     arr[0] = temp  
6     return arr
```

Array - Functions (Insert)

```
1 def insertElement(arr, size, elem, index):  
2     # Practice how to throw exception if there is no empty space  
3     if size == len(arr):  
4         print("No space left. Insertion failed")  
5     else:  
6         for i in range(size, index, -1):  
7             arr[i] = arr[i - 1] #Shifting right till the index  
8         arr[index] = elem #Inserting element  
9     return arr
```

Array - Functions (Remove)

```
1 def removeElement(arr, index, size):  
2     for i in range(index + 1, size):  
3         arr[i - 1] = arr[i] #Shifting left from removing index  
4     arr[size - 1] = None #Making last space empty
```